Lecture 1

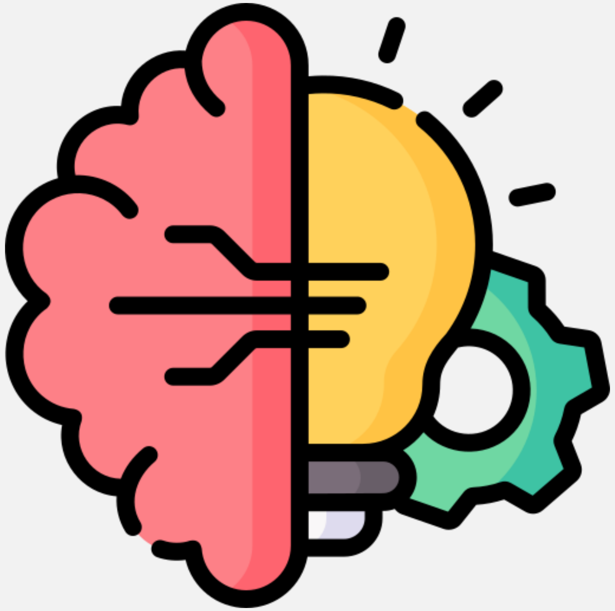
Introduction of Artificial Intelligence

Course overview

- **Learning Objectives**
  - Introduce and familiarise you with the Artificial Intelligence
  - Building conceptual understanding about ML, DL, AI and practically/theoretically distinguish them
  - Understand different types of ML and the corresponding real-world problems.
  - Build understanding about loss functions pertaining to Linear regression model.

Table’s Contents

- What is Artificial Intelligence
- AI vs ML vs DL
- Structured vs unstructured data
- Types of Machine Learning
- Regression vs Classification
- Loss Functions & Learning Objectives
- Linear Models – Linear Regression

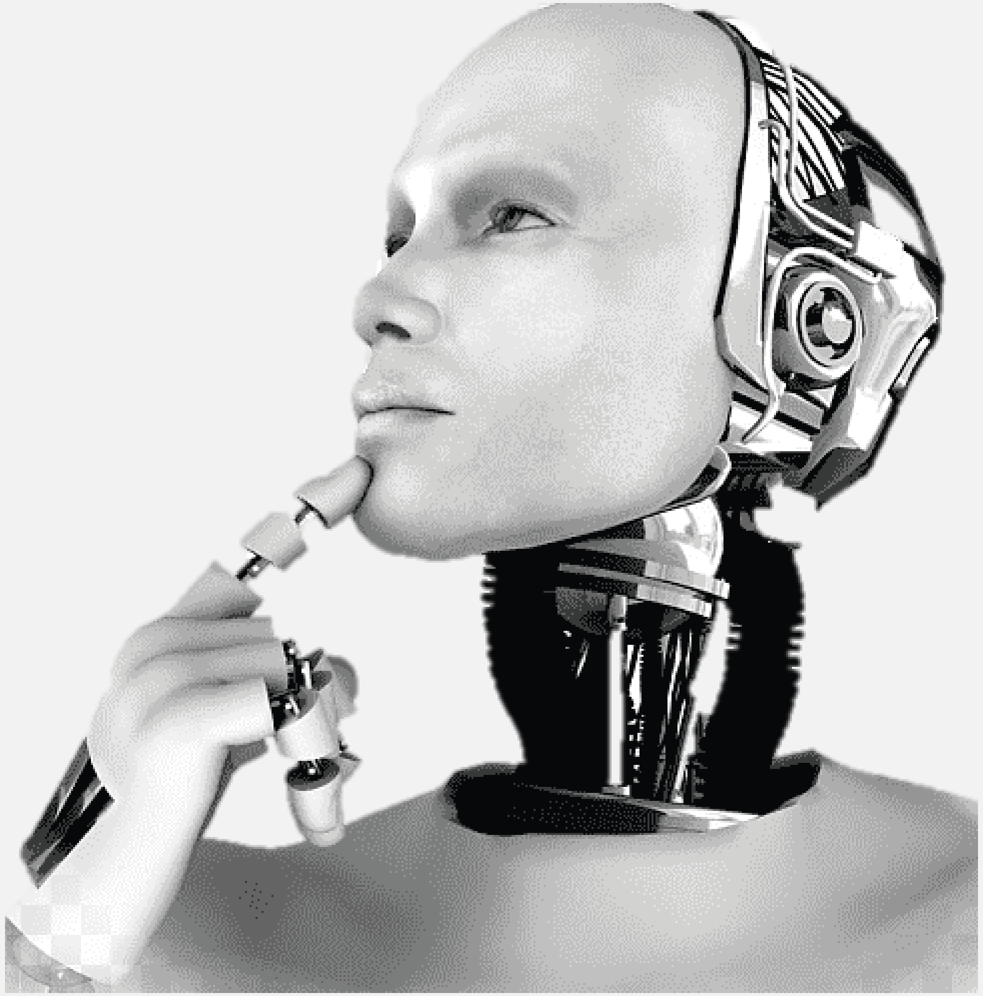


Note: Some of slides have been directly considered from Prof. Naeem Ullah Khan et al. lectures

What is Artificial Intelligence

Making the machine able to:

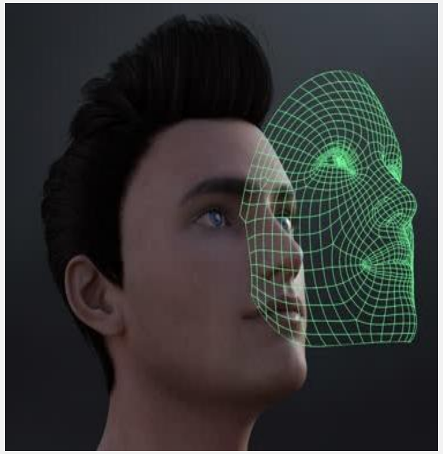
- Think like human
- Understand like human
- Behave like human



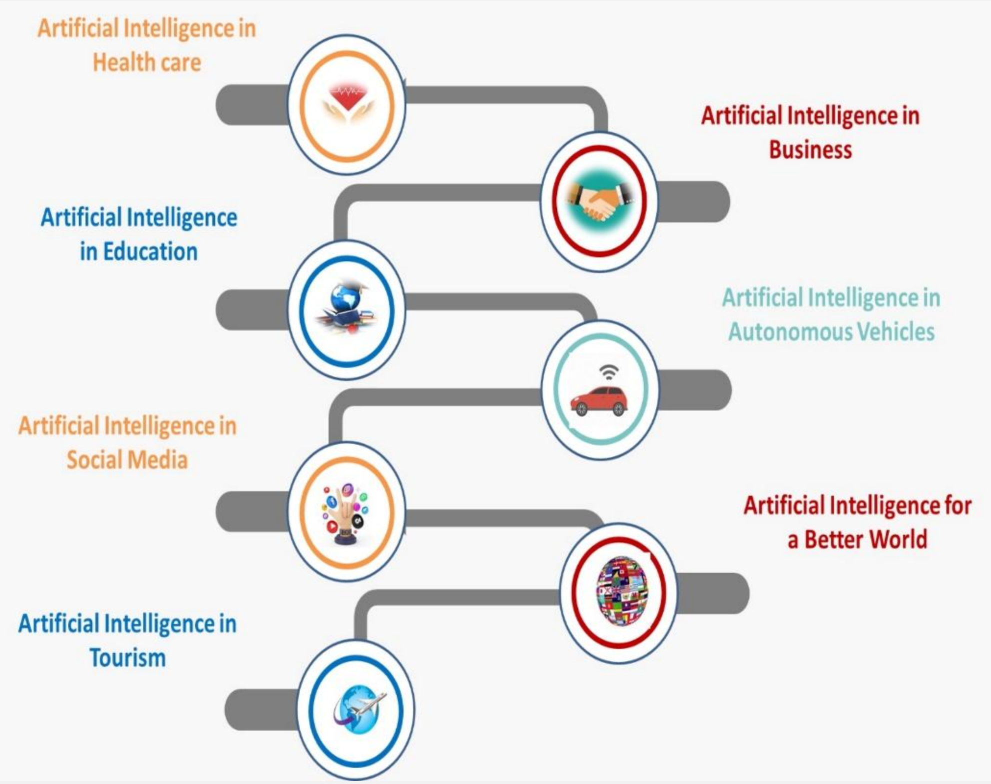
What is Artificial Intelligence

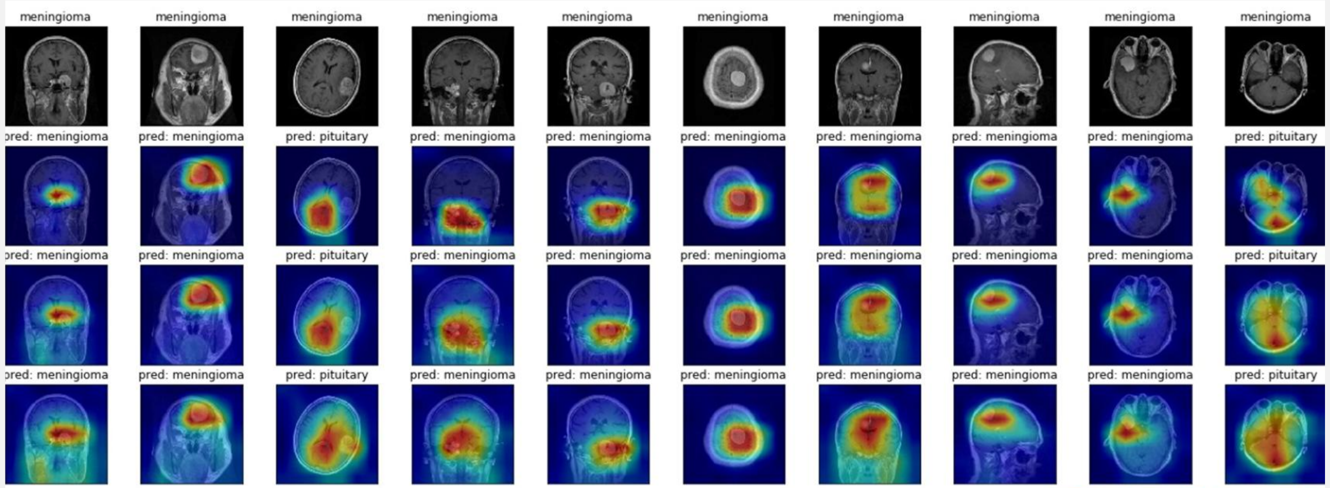
AI Is Not Similar To Human Intelligence. Thinking So Could Be Dangerous. [Elizabeth Fernande et al.]

Activity recognition, Object detection, Face recognition.

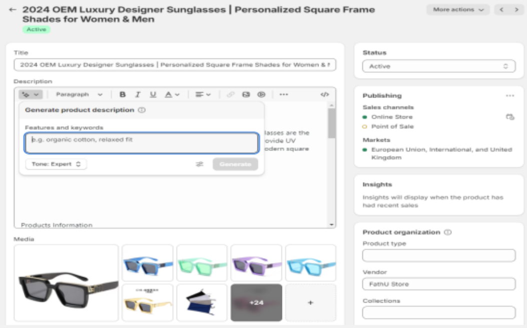


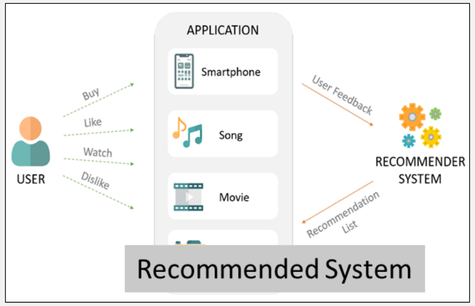
Applications of Artificial Intelligence





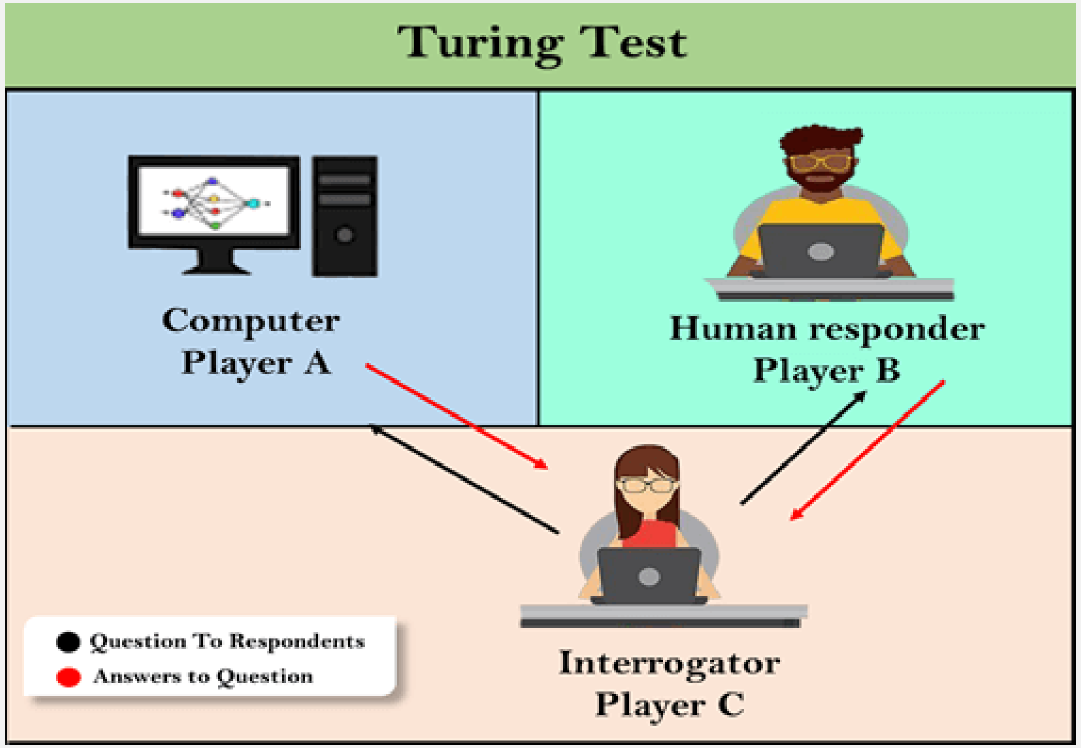
AI is designed to help diagnose and treat complex diseases such as brain tumors by combining technologies such as big data analytics, machine learning, and deep learning. AI can detect and classify tumors by analyzing brain imaging techniques, such as Magnetic Resonance Imaging MRI.





Acting Humanly: Turing Test Approach

Designed by Alan Turing (1950) as thought sidestepping the vagueness of the question “Can a machine think?”



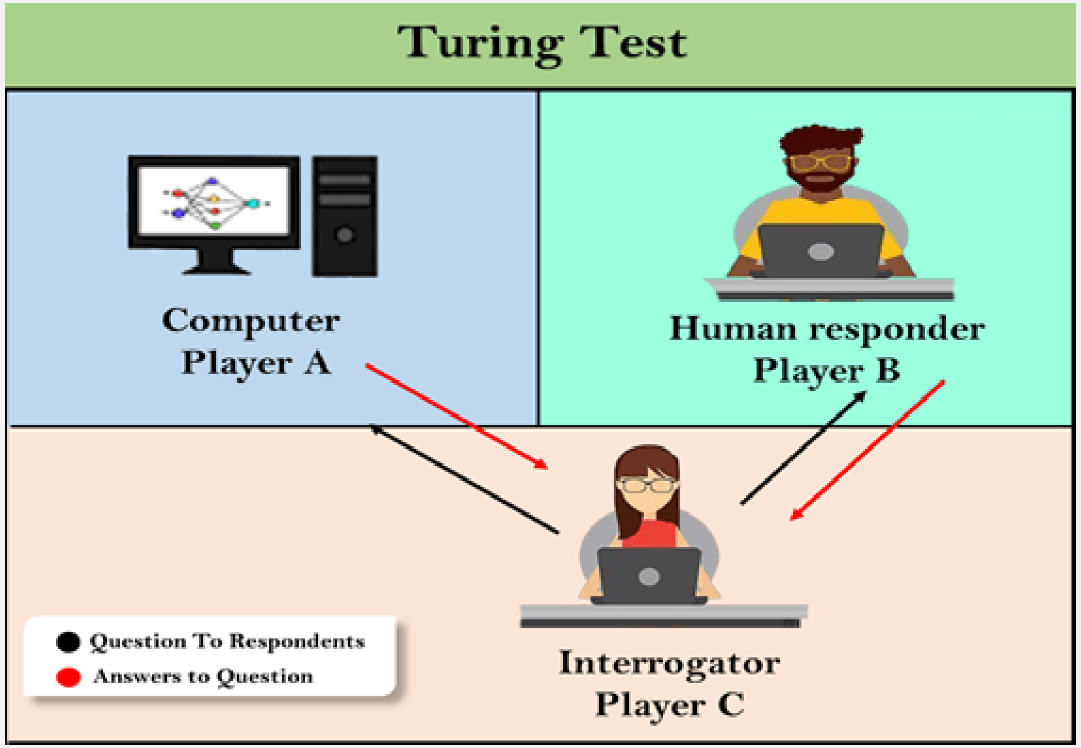
Machine would need:

- Natural language processing Communicate in human language
- Knowledge representation To store what it knows or hears;
- Automated reasoning Answer questions and draw new conclusions;
- Machine learning adapt to new circumstances, detect patterns

How it started?

Acting Humanly: Turing Test Approach

Designed by Alan Turing (1950) as thought sidestepping the vagueness of the question “Can a machine think?”



Machine would need:

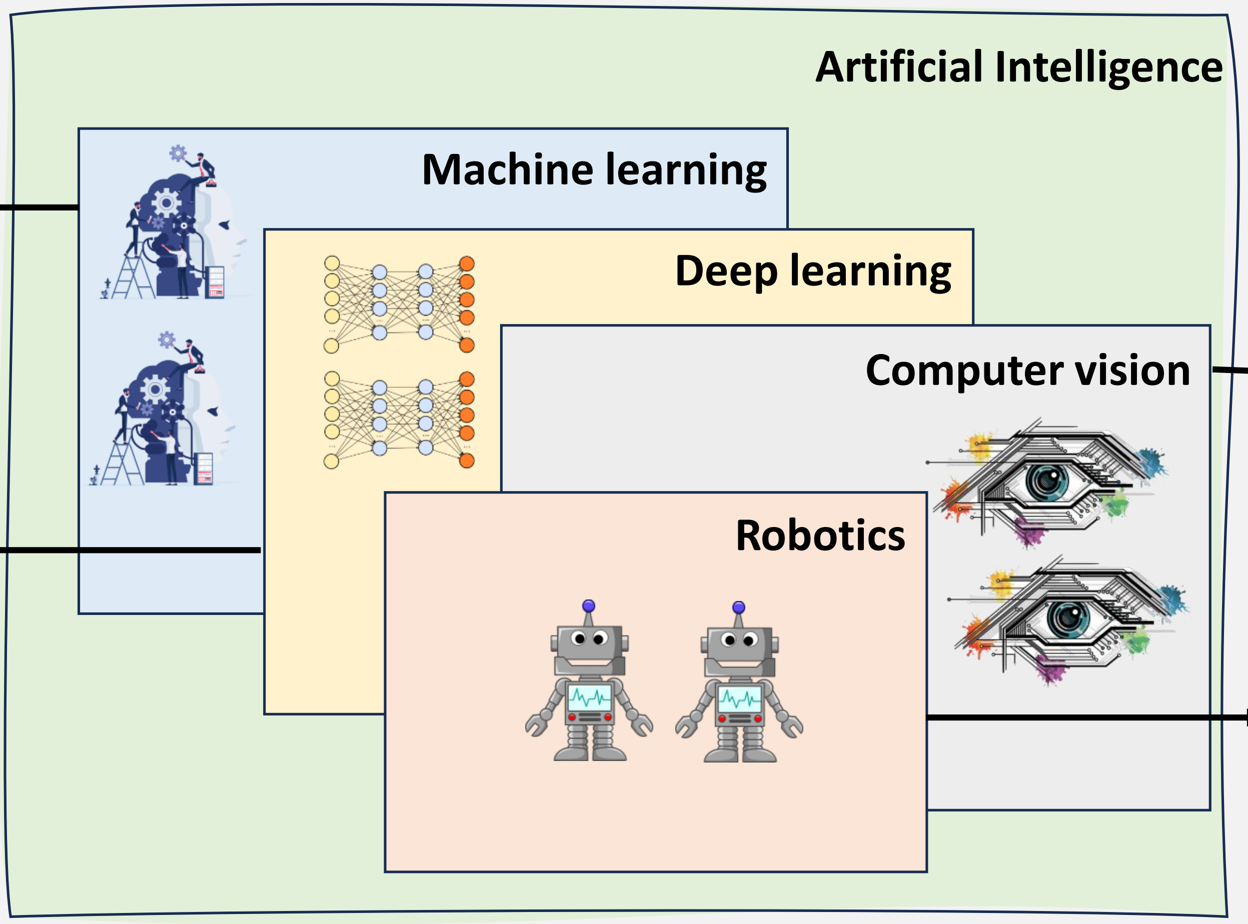
- Computer Vision and speech to perceive
- Robotics to manipulate object and move

These six disciplines compose most of AI.

Broader Picture of the Umbrella

- Supervised learning
- Unsupervised learning
- Reinforcement learning

- CNN, ConvNet, ANN
- RNN, LSTM, GRU
- Autoencoder



- Image processing, recognition
- Video analysis
- Image classification

- UAV, Drones
- Automation
- Cobots

AI vs ML vs DL

Machine Learning vs Deep Learning

أكاديمية كاوست
KAUST ACADEMY

1. Machine Learning (ML)

Algorithms that parse data, learn patterns from it, and apply learned decisions.

Characteristics:

- Best for **Structured Data** (Tables).
- Require **Feature Engineering**.
- Lightweight & Interpretable (mostly).

Examples: *Linear Reg, SVM, Trees*

2. Deep Learning (DL)

A subset of Machine Learning based on artificial neural networks.

Characteristics:

- Best for **Unstructured Data** (Images, Text, Audio).
- **Automated** Feature Extraction.
- Data & Compute Hungry.

Examples: *MLPs, CNNs, Transformers*

ML vs DL

Machine Learning ML

- What is Machine Learning?

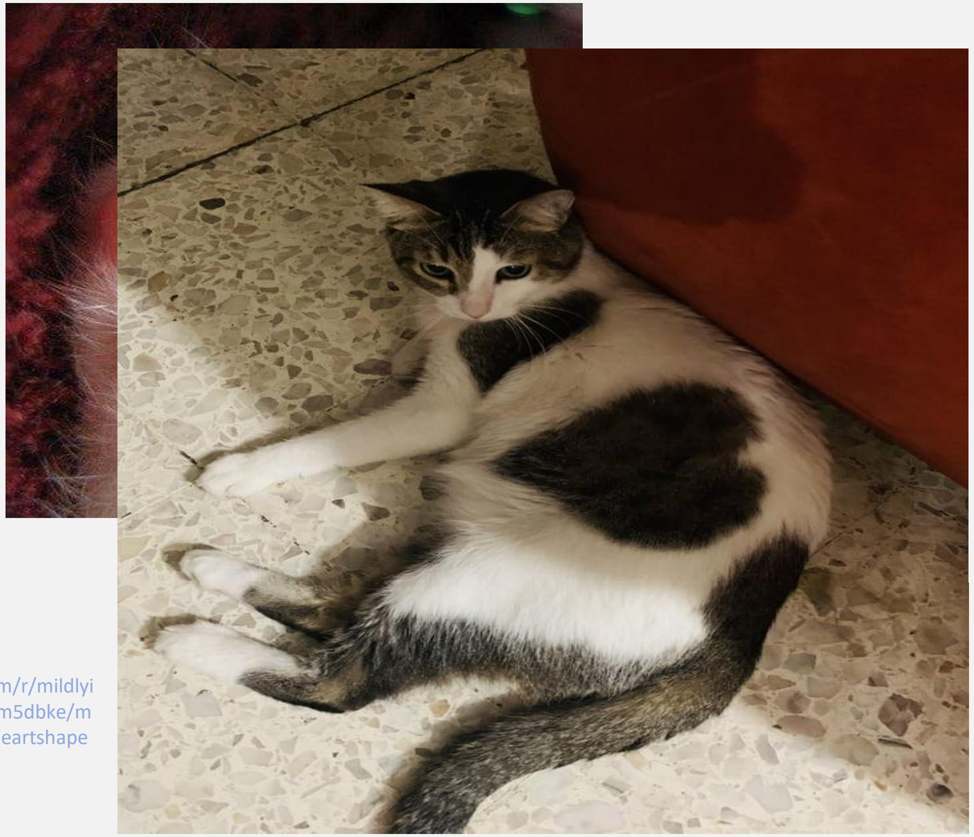
https://images.google.com/

Machine Learning ML

- What is Machine Learning?

https://www.amazon.co.uk/Photographic-Wallpaper-fotomural-Measures-Decoration/dp/B078KLW9LR

https://theprocrastinationpen.co.uk/2024/11/11/sleeping-with-asmr-92/

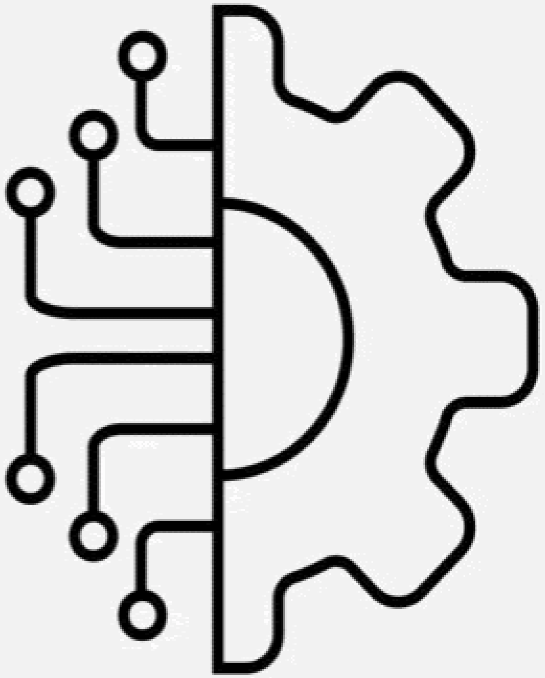


Irregularities in Shape

https://www.reddit.com/r/mildlyinteresting/comments/m5dbke/my_cat_has_a_perfect_heartshaped_spot_on_his_belly/

Machine Learning ML

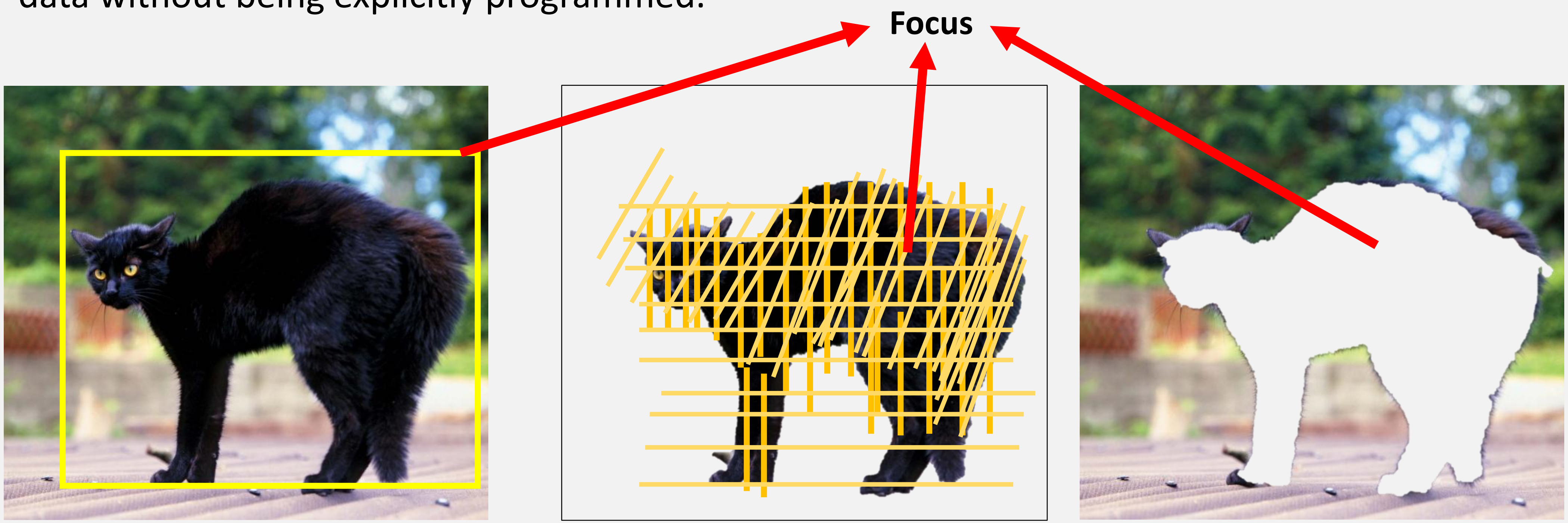
- What is Machine Learning?



Machine Learning ML

- What is Machine Learning?

Subfield of AI that focuses on giving machines the ability to learn and make decisions from data without being explicitly programmed.



Data Types

1. Structured Data

Organized in rows and columns.

- **Tabular Data**
  - Spreadsheets, SQL Databases
  - *Columns = Features, Rows = Samples*
- **Time-Series Data**
  - Data indexed by time
  - Stock prices, Weather, IoT sensors

2. Unstructured Data

Complex formats.

- **Text Data (NLP)**
  - Emails, Tweets, Documents
- **Images & Video (Computer Vision)**
  - Medical imaging, Surveillance, Face ID
- **Audio Data**
  - Voice commands, Music processing

Note: *Machine Learning excels at Structured Data, while Deep Learning dominates Unstructured Data.*

Types of Machine Learning

أكاديمية كاوست
KAUST ACADEMY

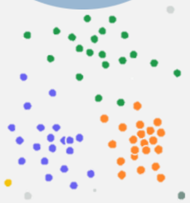
Machine Learning

1

Supervised Learning

2

Unsupervised Learning



3

Reinforcement Learning

Types of Machine Learning

Machine Learning

1

Supervised Learning

- Model training is performed with Label data
- Learning algorithm is provided with inputs and the corresponding correct outputs (labels).

Types of Machine Learning

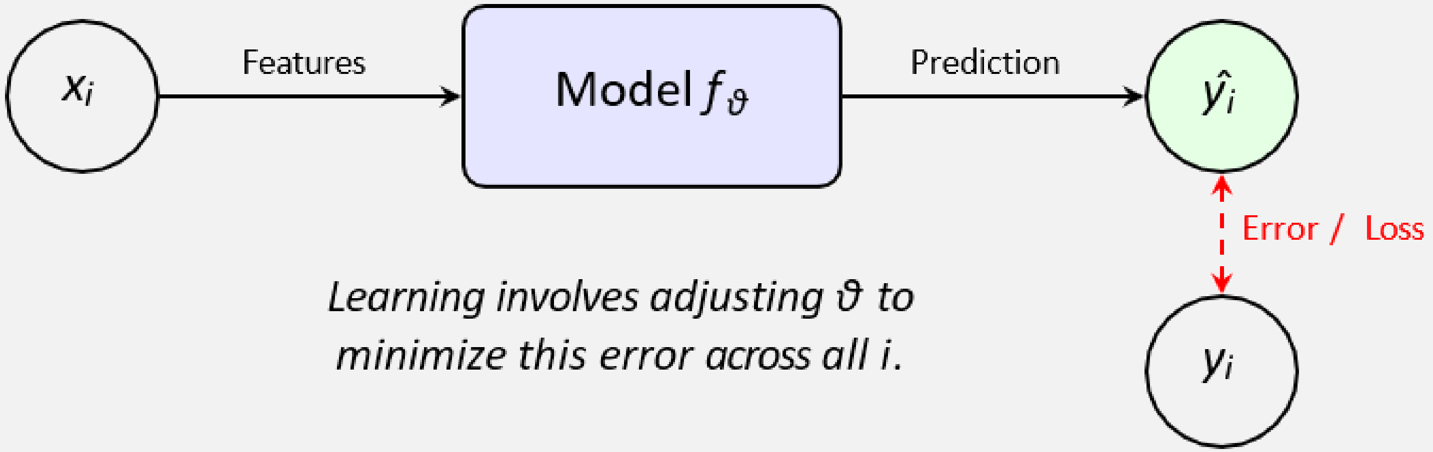
1 Supervised Learning

The Objective

Given a labeled dataset $D = \{(x_i, y_i)\}_{i=1}^{n}$, we aim to learn a mapping function f that generalizes to unseen data:

$$\hat{y} = f(x;\vartheta) \approx y$$

where ϑ represents the learnable model parameters.



Learning involves adjusting ϑ to minimize this error across all i.

Types of Machine Learning (Supervised Learning)

Example 1: Binary Classification (*Cat* & *Dog*)

Definition:

Target y is **discrete**: $y \in \{0, 1\}$ (e.g., spam/not spam)

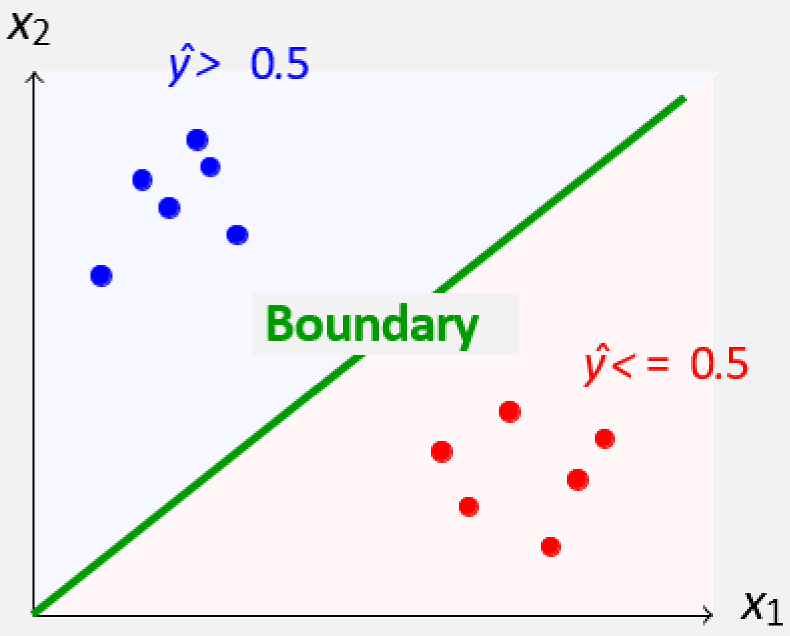
How It Works:

1. Model outputs **probability**: $\hat{y} \in [0, 1]$
2. Apply **threshold** (usually 0.5) to classify
3. If $\hat{y} \geq 0.5 \rightarrow$ Class 1, else Class 0

Example

$\hat{y} = 0.87 \xrightarrow{\text{threshold } 0.5}$ **Class 1** (Spam)

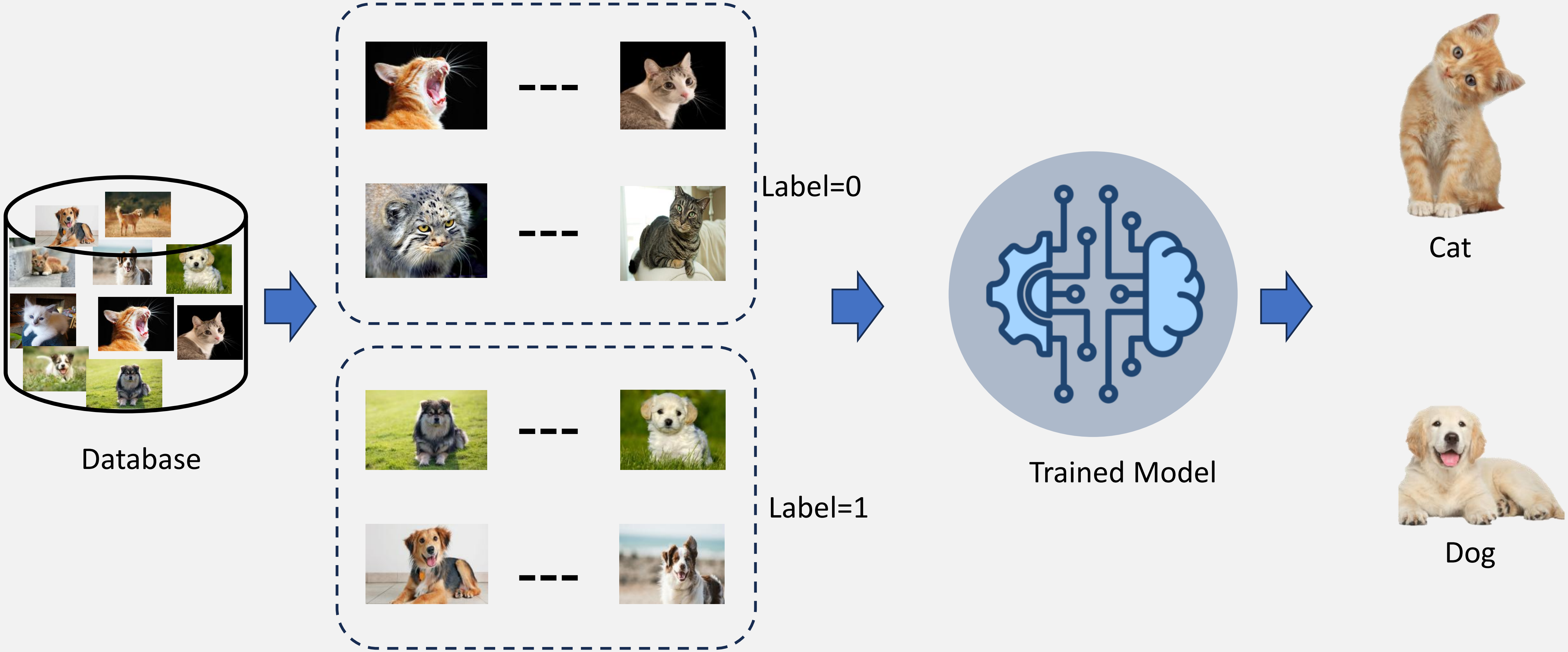
Decision Boundary Visualization



Goal: Find optimal boundary that maximizes class separation

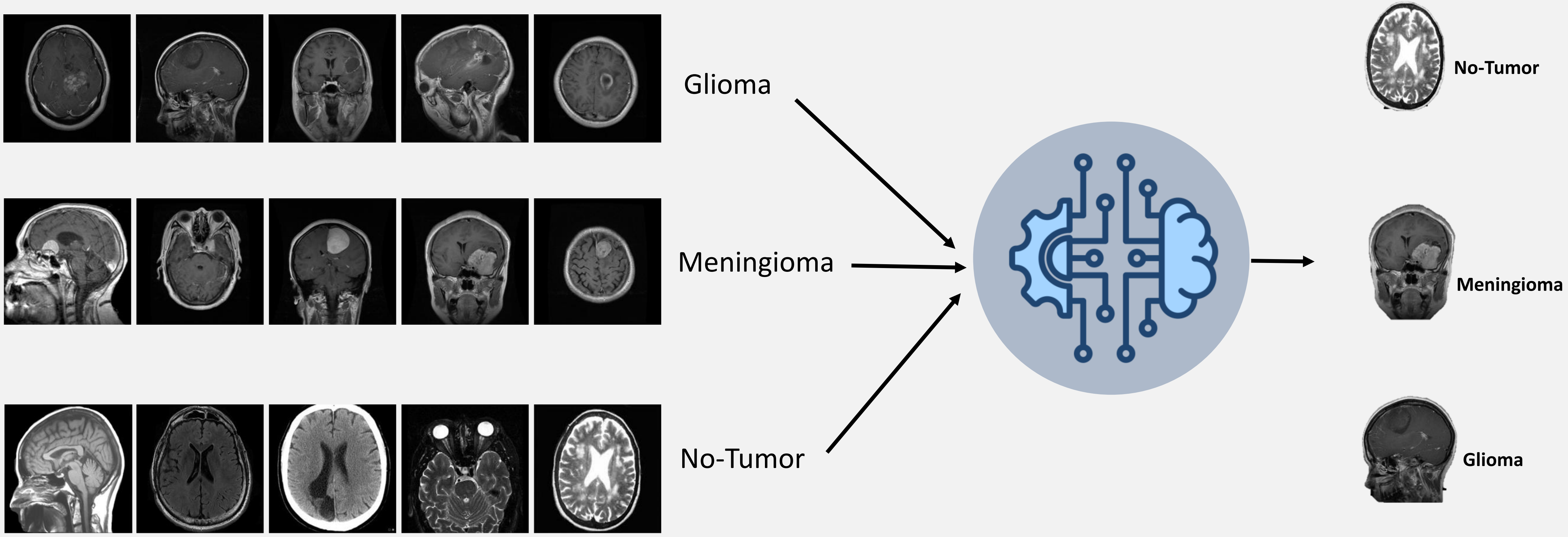
Types of Machine Learning (Supervised Learning)

Example 1: Binary Classification (*Cat* & *Dog*)



Types of Machine Learning (Supervised Learning)

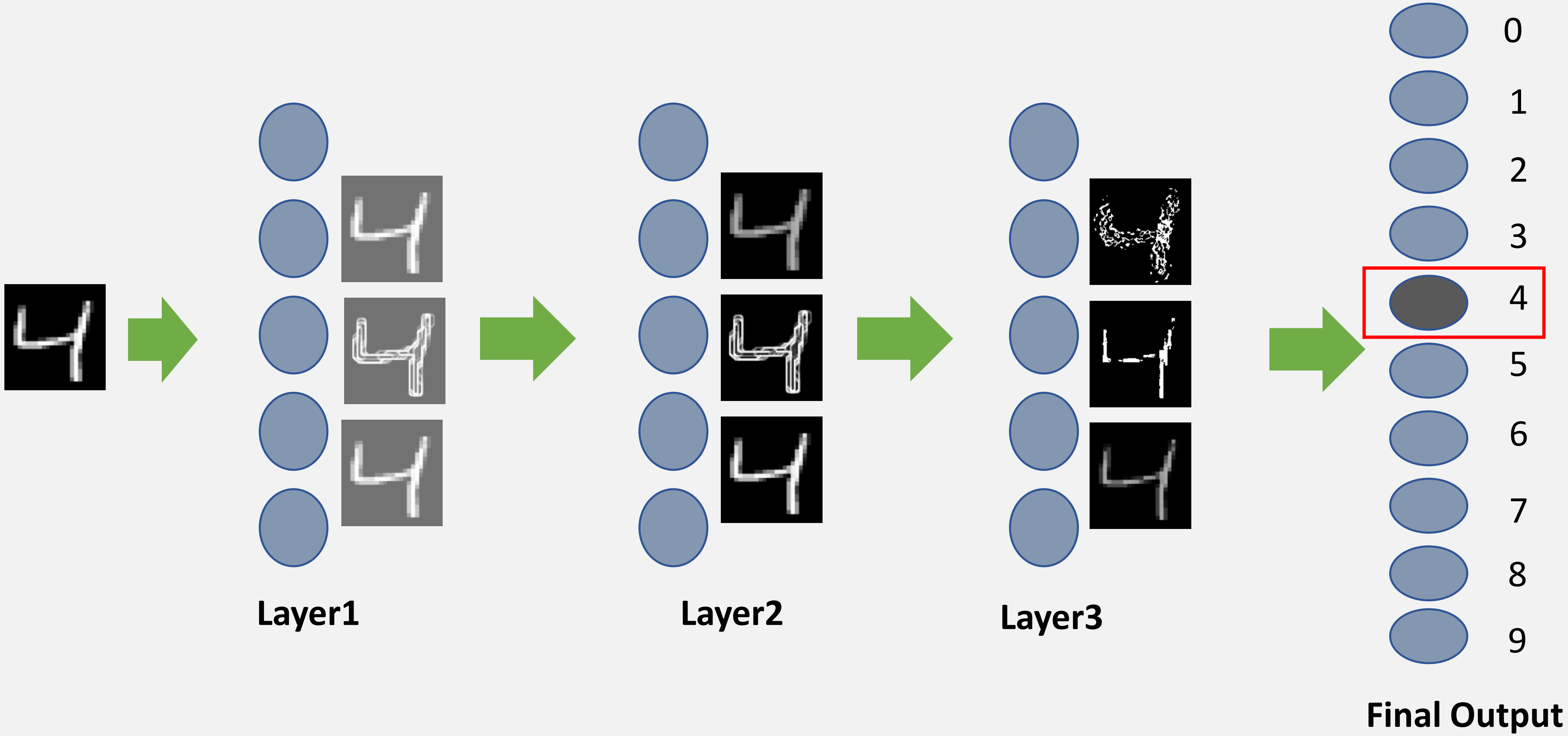
Example 2: Multi-classification (*Brain Tumors*)



https://www.mdpi.com/2079-9292/14/4/710

Types of Machine Learning (Supervised Learning)

Example 3: Text Classification (*digits classification*): DIGITS dataset



Types of Machine Learning (Supervised Learning)

Example 4: Regression

Definition: The target y is **Continuous** ($y \in \mathbb{R}$).

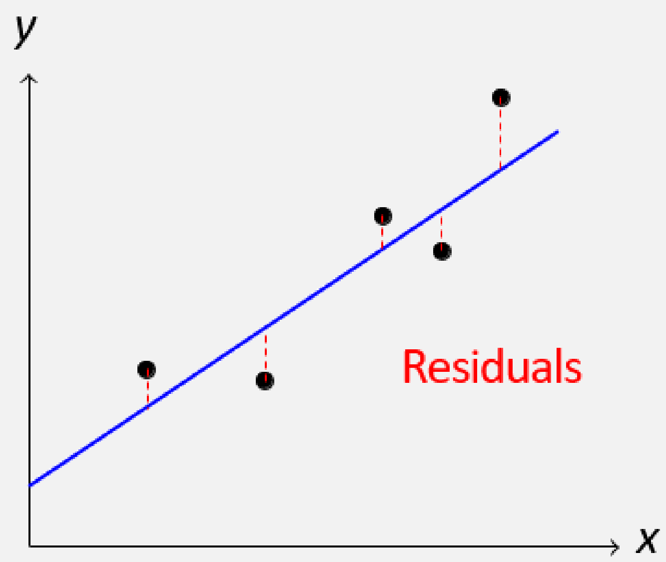
Examples:

- House Prices (\$300k, \$500k)
- Temperature (24.5°C)
- Stock Value

The Loss: Mean Squared Error

Measures the squared distance between points and the line.

$$MSE = \frac{1}{N}\sum (y - \hat{y})^2$$



Goal: Minimize the sum of red lines.

Coming ahead in detail

Algorithm in Supervised Learning

- **Linear Regression:** Predicts continuous outcomes.
- **Logistic Regression:** Used for binary classification tasks.
- **Support Vector Machines (SVM):** Finds the hyperplane that best separates different classes.
- **Decision Trees:** Tree-like models used for both classification and regression tasks.

Types of Machine Learning

Machine Learning

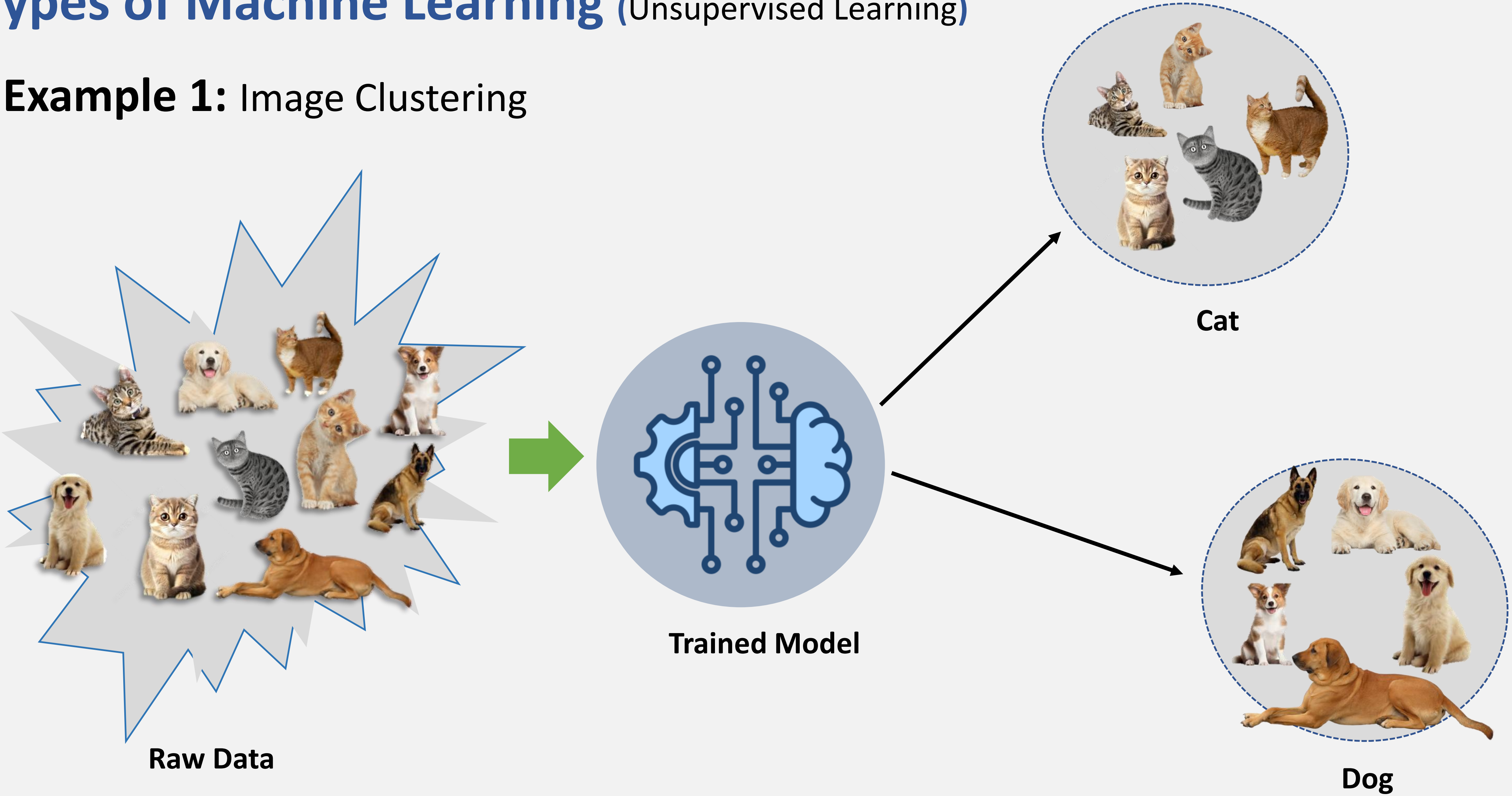
2

Unsupervised Learning

- Model training is performed without Label data
- Learning algorithm finds patterns/structure/features in the data

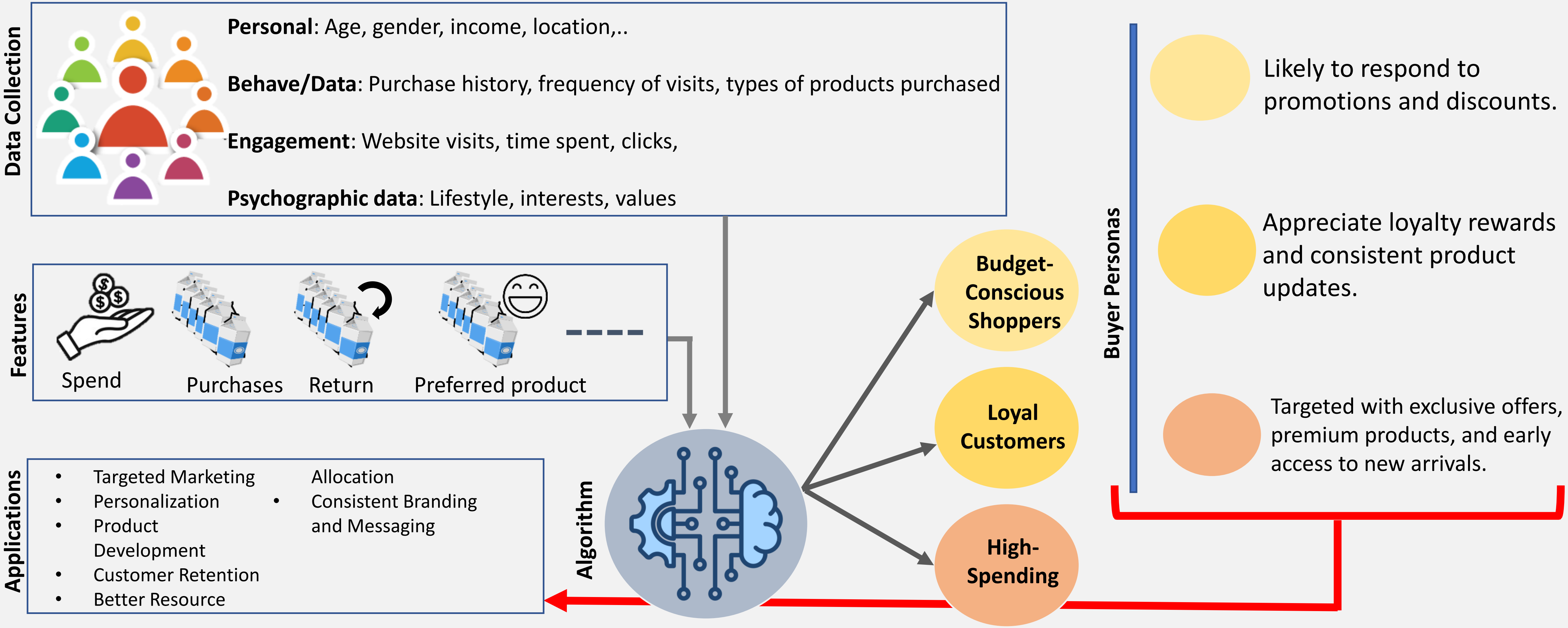
Types of Machine Learning (Unsupervised Learning)

Example 1: Image Clustering



Types of Machine Learning (Unsupervised Learning)

Example 2: Customer Segmentation



Types of Machine Learning (Unsupervised Learning)

Algorithms in Unsupervised Learning

- **K-means cluster:** Divides data into K clusters, each data point belongs to the cluster with the nearest mean..
- **Principal Component Analysis (PCA):** Reduces the data dimensionality by finding the main components that capture the most variance..

Types of Machine Learning

Machine Learning

3

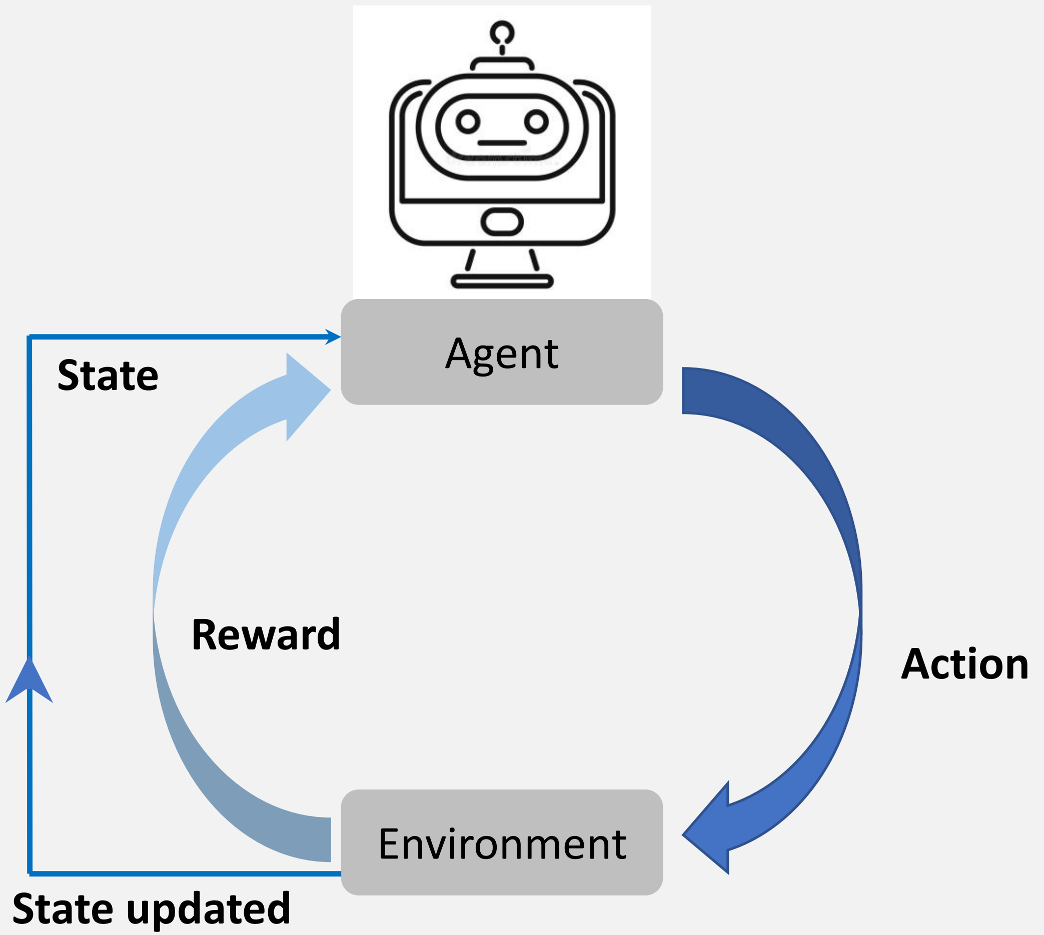
Reinforcement Learning

- Involves an agent that learns to interact with its environment to maximize a reward signal.
- The agent takes actions, receives feedback (reward / punishment), and adjusts its behavior accordingly.

Types of Machine Learning (Reinforcement Learning)

Key Concepts in RL

- **Agent:** Decision maker/taker.
- **Environment:** Space the agent interacts with.
- **Rewards**: Feedbacks/points for positive/good actions.
- **Policy**: Strategy the agent learns for choosing actions.



Supervised Learning models

We will start with supervise learning models (regression and classification) and implementation for real world problems.

Though, going forward the relevant terminologies will be covered including Loss functions, core ML challenges, and regularization techniques.

Loss Functions

It measures how far a model's predictions are from the true values. During training, the model adjusts its parameters to minimize the loss, which is why the loss function is often called the learning objective. What is Loss Function? | IBM

Mean Squared Error (MSE) Loss Function

Mean Squared Error is the average of the squared differences between predicted values and actual values.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$$

- Smooth and differentiable.
- Easy to optimize with gradient descent.
- Strongly discourages large errors.

But
Very sensitive to outliers.
A few extreme values can dominate the loss

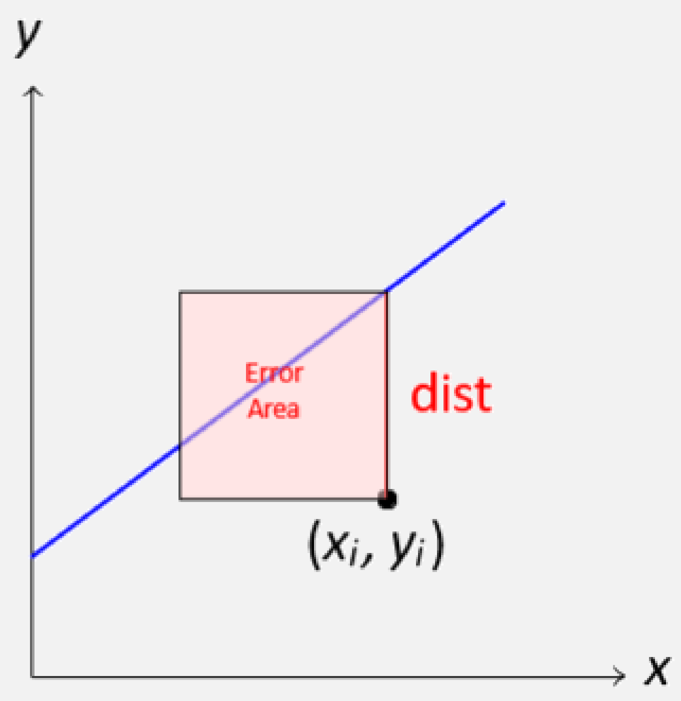
Mean Squared Error (MSE) Loss Function

Loss Functions

Why Squared Error?

The 3 Steps to MSE Loss:

1. **Residuals ($y - \hat{y}$):** measures the raw distance. Points below the line give negative distances.
2. **The Square $(\ldots)^2$:** Squaring ensures all distances are positive so they don't cancel out. It also heavily penalizes **outliers**.
3. **The Average ($\frac{1}{N}\sum$):** We aggregate errors from all N samples to get a single performance score.



Each "error" acts as a square area we want to shrink to zero.

Loss Functions

It measures how far a model's predictions are from the true values. During training, the model adjusts its parameters to minimize the loss, which is why the loss function is often called the learning objective. What is Loss Function? | IBM

Mean Absolute Error (MAE) Loss Function

Mean Absolute Error computes the average absolute difference between predictions and targets.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} | x_i - x |$$

Loss Functions and Metrics in Deep Learning

Robust to noisy data.

Less affected by extreme values.

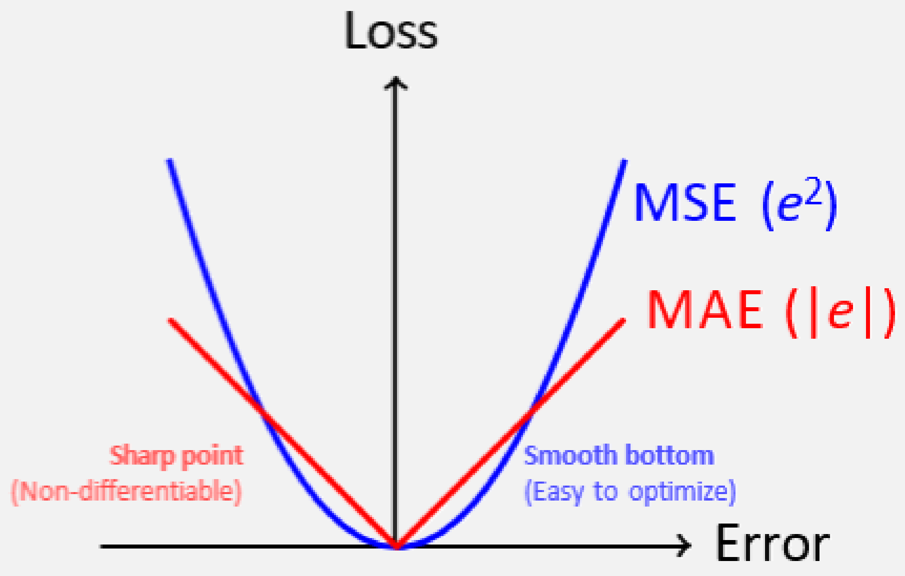
Easy to interpret.

Loss Functions

Why Squared Error?

Note

If we use the **Absolute** difference $|y - \hat{y}|$ in-stead of the square, we get **MAE**. It is more robust to outliers, but **not differentiable** at 0.
Mathematically, you can still optimize functions with **finitely** many non-differentiable points using subgradients, though this adds some complexity.



MSE (Blue) is smooth. MAE (Red) is sharp.

Loss Functions

Why Squared Error?

Problem with MSE

MSE measures **numeric distance**, but in classification, we care about **probability confidence**, not distance!

Loss Functions

Why Squared Error?

Problem with MSE

MSE measures **numeric distance**, but in classification, we care about **probability confidence**, not distance!

What We Really Want

A loss function that **heavily penalizes confident wrong predictions**

Example: Classifying an image as a dog with 98% confidence when it's actually a cat should incur massive penalty!

Loss Functions

Why Squared Error?

Problem with MSE

MSE measures **numeric distance**, but in classification, we care about **probability confidence**, not distance!

What We Really Want

A loss function that **heavily penalizes confident wrong predictions**

Example: Classifying an image as a dog with 98% confidence when it's actually a cat should incur massive penalty!

Solution: The Logarithm!

The logarithm function has exactly the property we need:
it grows exponentially as confidence in the **wrong** class increases (**This will continue on Day3**)

The ML Toolbox: Types of Models

Machine Learning Models (f_ϑ)

1. Linear Models
Linear Regression
Logistic Regression

2. Distance-Based
KNN
(Non-Parametric)

3. Kernel-Based
SVM

4. Tree-Based
Decision Trees
Random Forest
Gradient Boosting

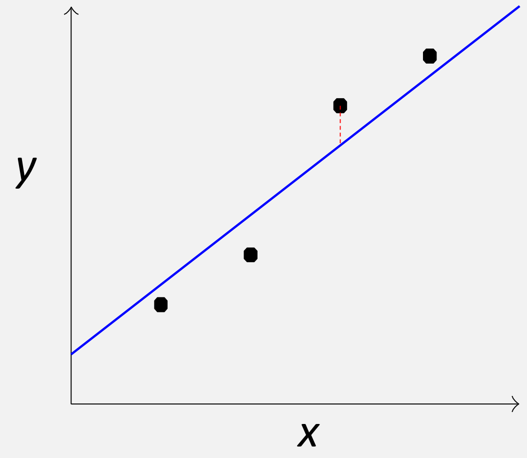
↑ **We Start Here!**

Supervised Learning Algorithms – Regression

Simple Linear Regression

- Model the relationship between a dependent variable and one or more independent variables.

Goal: Find the best-fitting line (or hyperplane in higher dimensions) that predicts the dependent variable based on the independent variable(s).



Dependent (Target): Outcome we want to predict (e.g., house prices)

Independent (Predictor): Variable used to make predictions (e.g., Square foot, No. of bedrooms, etc.)

Supervised Learning Algorithms – Regression

Simple Linear Regression

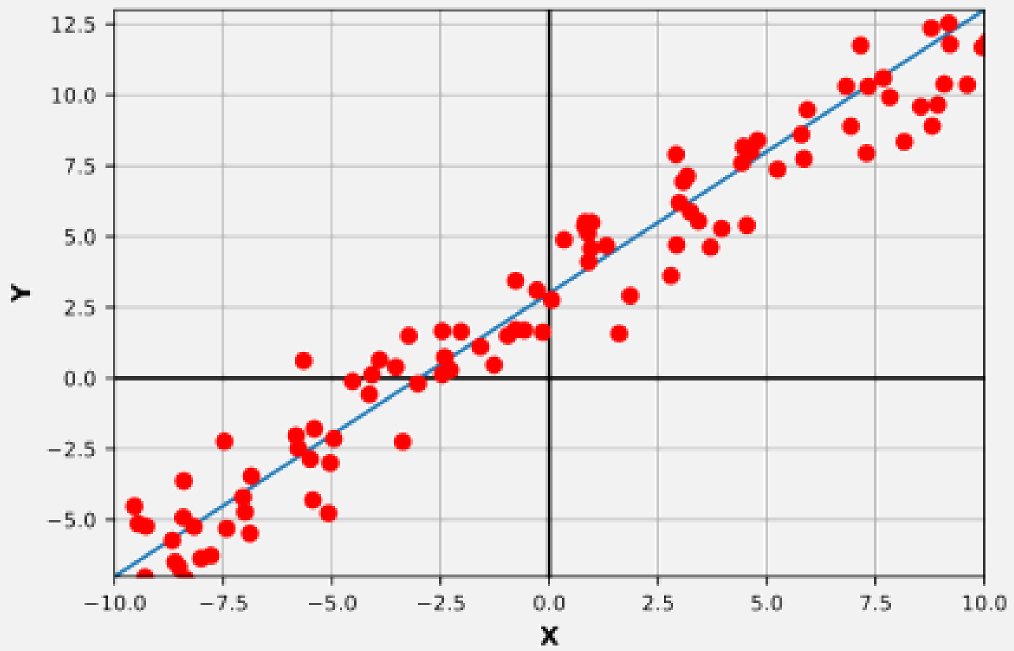
y = mx + b

:**y** is the dependent variable.

:**m** is the slope of the line (change in y for a unit change in x).

:**x** is the independent variable.

:**b** is the y-intercept (value of y when x=0).



Supervised Learning Algorithms – Regression

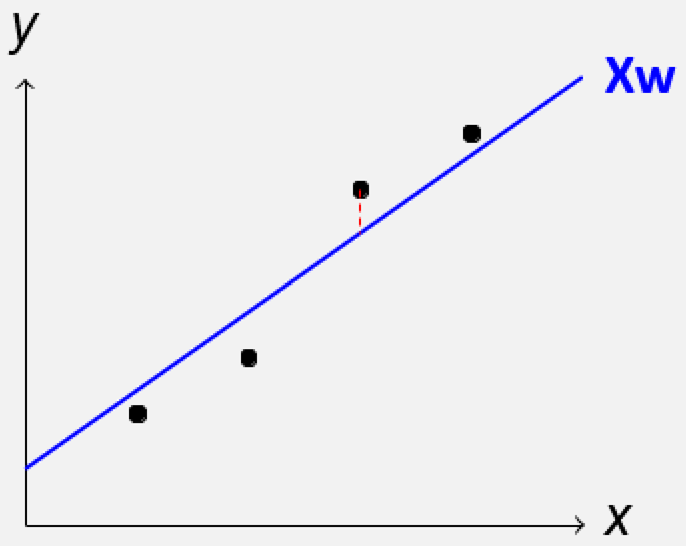
Simple Linear Regression

We start with the simplest assumption:
The target y is a **linear combination** of the input features.

$$\hat{y} = w_0 + w_1x_1 + w_2x_2 + \cdots + w_dx_d$$

The Components:

- $\hat{y}$: predicted value
- $x_1, \ldots, x_d$: the input features.
- $w_1, \ldots, w_d$: the weights. How much each feature matters.
- w_0: the bias (Intercept).



Writing long sums ($\sum$) is tedious. We can use Linear Algebra to represent Linear Regression in matrices notation.

Supervised Learning Algorithms – Regression

Translating to Vectors (Vectorization)

Step 1: The Bias Trick: We add a "dummy" feature $x_0 = 1$ to the input so we have each weight multiplied by a x feature.

$$\hat{y} = w_0(1) + w_1 x_1 + \cdots + w_d x_d$$

Step 2: Vector Notation (For one sample): We rewrite this equation in a vector dot product form:

$$\hat{y} = \begin{bmatrix} w_0 & w_1 & \dots & w_d \end{bmatrix} \cdot \begin{bmatrix} 1 \\ x_1 \\ \vdots \\ x_d \end{bmatrix} = \mathbf{w}^T \mathbf{x}$$

Step 3: Matrix Notation (For all data): For N samples, prediction becomes a simple Matrix-Vector multiplication:

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{w}$$

Supervised Learning Algorithms – Regression

Finding the Best Line

Now we have the equation. But how do we choose the best **w**?

Supervised Learning Algorithms – Regression

Finding the Best Line

Now we have the equation. But how do we choose the best **w**?

We need the loss and the optimizer!

Supervised Learning Algorithms – Regression

Finding the Best Line

Now we have the equation. But how do we choose the best **w**?

We need the loss and the optimizer!

Let's wrap our model in the Mean Squared Error:

$$J(\mathbf{w}) = \frac{1}{N}||\mathbf{X}\mathbf{w} - \mathbf{y}||_2^2$$

Supervised Learning Algorithms – Regression

Finding the Best Line

Now we have the equation. But how do we choose the best **w**?

We need the loss and the optimizer!

Let's wrap our model in the Mean Squared Error:

$$J(\mathbf{w}) = \frac{1}{N}||\mathbf{X}\mathbf{w} - \mathbf{y}||_2^2$$

Now, how do we find the **w** that minimizes this $J(\mathbf{w})$?

Supervised Learning Algorithms – Regression

Finding the Best Line

Now we have the equation. But how do we choose the best **w**?

We need the loss and the optimizer!

Let’s wrap our model in the Mean Squared Error:

$$J(\mathbf{w}) = \frac{1}{N}||\mathbf{X}\mathbf{w} - \mathbf{y}||_2^2$$

Now, how do we find the **w** that minimizes this *J*(**w**)?

Option A: Analytical
(Solve for zero gradient)

OR

Option B: Itérative
(Gradient Descent)

Supervised Learning Algorithms – Regression

Method A: The Normal Equation

For linear models, MSE is **convex**. **Q:** What does that tell you? *Guaranteed Global Minimum*!.

Step 1: The Gradient Derived via Chain Rule:

$$\nabla J(\mathbf{w}) = \frac{2}{N}\mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y})$$

Step 2: Set to Zero

$$\mathbf{X}^T\mathbf{X}\mathbf{w} - \mathbf{X}^T\mathbf{y} = 0$$

$$\mathbf{X}^T\mathbf{X}\mathbf{w} = \mathbf{X}^T\mathbf{y}$$

Step 3: Solve for w Multiply both sides by inverse $(\mathbf{X}^T\mathbf{X})^{-1}$:

The Normal Equation

$$\mathbf{w} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Cons: Inverting large matrices is very slow $(O(d^3))$. Doesn't work if X is singular.

Supervised Learning Algorithms – Regression

Method B: Gradient Descent

Instead of jumping to the solution, we take small steps.

Step 1: Calculate Gradient Derived via Chain Rule:

$$\nabla J(\mathbf{w}) = \frac{2}{N}\mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y})$$

Step 2: Update Rule Move opposite to the gradient to reduce error.

Iterative Update

$$\mathbf{w}_{new} = \mathbf{w}_{old} - \alpha \nabla J$$

α: Learning Rate (Step size)

Pros: Very efficient for large datasets.

Any Question?